

At

Home of Computer

my Savings

That's why I'm doing this

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EXAM → Practical & Theory

Scheduling

1. First come first serve

It's divided into 2 mainly

1. Process with same priority

2. Priority

3. Process with different priority

For the above processes the time

a For Processes whose arrival are the same
Process, Burst time will be given

You'll have to calculate Waiting Time and Turn around time
, 'Total average of waiting time and turn around time'
It could be, "Write the code for it".

b For Processes, where arrival are not the same

Process, Arrival Time and Burst time will be given
You are to find 'Completion time', 'Inter arrival time',
'Waiting time', 'Turn around time', you can also be
asked to find 'Average time of Completion, Inter Arrival, Waiting
and Turn around time'

'It could be in form of practical or theory'
So, you'll calculate or write a code

1.2

a	Process	burst time	waiting time	Turnaround	Average waiting time
	P ₁	5	0	5	$\frac{0+5+12}{3} = 5.667$
	P ₂	7	5	12	
	P ₃	6	12	18	

Average Turn around time

$$\frac{5+12+18}{3} = 11.667$$

b	Process	Arrival time	Burst time	Completion time	Interarrival Time	Waiting time
	P ₁	0	5	5	0	0
	P ₂	5	7	12	5 (5-0)	0
	P ₃	10	6	18	5 (10-5)	2

* Completion time $\geq 5, 5+7,$ Turn around Time

Interarrival $\geq$ The gap between Process 1 and Process 2

So 0, 5-0, 10-5, 6+7+5

5

7

8

? * Waiting time $\rightarrow$? $\rightarrow$ calculate Turn around time from

* Completion Time - Arrival Time $\geq$ Turn around time

$$5 - 0 \geq 5$$

$$12 - 5 \geq 7$$

$$18 - 10 \geq 8$$

* Waiting Time $\geq$ Turn around time - Burst Time

$$5 - 5 \geq 0$$

$$7 - 7 \geq 0$$

$$8 - 6 \geq 2$$

Average Completion time = $\frac{5+12+18}{3}$

Average Inter Arrival time

3

Average waiting time

For all three,
just calculate
add up all the
numbers in brackets
and then
the answer is
number of processes

Coding

Cont'd of Coding: We are adding Turn around now - NB -
under $wt[i] = wt[i] + bt[j]$ it could be line 22.

Define turn around time

Line 4 `int n, i, j, bt[10], wt[10], tat[10]`

Line 22 `for (i=0; i<n; i++) {`

`tat[i] = bt[i] + wt[i];`

// now add it to line 29

Line 29 `printf("process burst time waiting time turn around time \n")`

`for (i=0; i<n; i++) {`

`printf("P[%d] \t %d \t %d \t %d \n",`

`i+1, bt[i], wt[i], tat[i]);`

`}`

// now to calculate the average, Declaring a float

After Line 4, `int n, i, j, ...`

5 `float atat = 0, awb = 0`

Line 23 `for (i=0; i<n; i++) {`

```
    tat[i] = bt[i] + wt[i];
```

```
    atat = atat + tat[i];
```

```
}
```

```
    atat = atat / n;
```

```
// now to calculate average waiting time
```

```
for (i = 0; i < n; i++) {
```

```
    awt = awt + wt[i];
```

```
}
```

```
    awt = awt / n;
```

```
// now go to the line before return to print the  
out
```

```
printf("average turn around time is: %0.3f\n", atat);
```

```
printf("average waiting time is : %0.2f\n", awt);
```

```
// now we want to give numbers to each processes
```

Just like the Operating system gives a number to every processor

```
// add these 2 headlines
```

```
1 #include <stdio.h>
```

```
2 #include <sys/types.h>
```

```
3 #include <unistd.h>
```

Using C++ (the process number at the same time)

with $P_1 = 5$, $P_2 = 7$, $P_3 = 6$

```
1 #include <stdio.h>
```

```
// under line 7 which is float atab = 0, qtab = 0;
```

```
Line 9 pid_t id;
```

```
id = getpid();
```

program is : %d

```
printf("The process id number for this program is : "\n"  
      (int)id)
```